LINUX USER AND ACCOUNT MANAGEMENT

Practical Lab Evidence & Progress Record

Programme	Google Cybersecurity Certificate
Provider / Platform	Google / Coursera
Lab activity	Manage user access in Linux
Status	Completed

1. Lab Overview

This document records the steps completed in the Linux user and account management lab. It provides space to add terminal screenshots as evidence of each task and its result.

The scenario follows the lifecycle of the Linux account researcher9: creating the account, assigning a primary group, transferring ownership of a project file, adding a supplementary group, and removing the account when access is no longer required.

2. Learning Objectives

- Create and manage a Linux user account.
- Assign a primary group and add a supplementary group.
- Change ownership of a file using chown.
- Remove a user account and clean up its associated group.
- Use sudo to perform administrative account-management operations.
- Relate account lifecycle management to identity and access management (IAM) and least privilege.

3. Lab Environment

Initial lab user: analyst

Initial working directory: /home/analyst

Employee account: researcher9

Primary group: research_team

Supplementary group: sales_team

Project file: /home/researcher2/projects/project_r.txt

Command requirement: The lab instructions require sudo at the beginning of each administrative command.

4. Task 1 — Add a New User

A new employee joins the Research department. Create the researcher9 account, then assign research_team as the account's primary group.

Step 1.1 — Create the user account

Evidence: Capture the terminal after running the command. If the command produces no output, show the command and the next prompt.

Screenshot 1 — researcher9 user account creation

```
analyst@3b53409cfca1:~$ sudo useradd researcher9
analyst@3b53409cfca1:~$ sudo usermod -g research_team researcher9
analyst@3b53409cfca1:~$
```

Step 1.2 — Assign the primary group

```
sudo usermod -g research_team researcher9
```

Evidence: Capture the command and, if you verify the account, include the verification output.

```
id researcher9
```

Screenshot 2 — research_team assigned as primary group

```
analyst@3b53409cfca1:~$ sudo usermod -g research_team researcher9
usermod: no changes
analyst@3b53409cfca1:~$ id researcher9
uid=1003(researcher9) gid=1003(research_team) groups=1003(research_team)
analyst@3b53409cfca1:~$
```

5. Task 2 — Assign File Ownership

Transfer ownership of project_r.txt to researcher9. The file is located in /home/researcher2/projects.

Evidence: Capture the command and verify the updated owner. For example:

Screenshot 3 — project_r.txt ownership changed to researcher9

```
analyst@3b53409cfca1:~$ sudo chown researcher9 /home/researcher2/projects/project_r.txt
analyst@3b53409cfca1:~$ ls -l /home/researcher2/projects/project_r.txt
-rw-rw-r-- 1 researcher9 research_team 46 Nov  5 02:55 /home/researcher2/projects/project_r.txt
analyst@3b53409cfca1:~$
```

6. Task 3 — Add a Supplementary Group

The employee now works across Research and Sales. Add sales_team as a supplementary group while retaining research_team as the primary group.

Evidence: Verify the user's primary and supplementary group memberships:

Screenshot 4 — sales_team added as a supplementary group

```
analyst@3b53409cfca1:~$ sudo usermod -a -G sales_team researcher9
analyst@3b53409cfca1:~$ id researcher9
uid=1003(researcher9) gid=1003(research_team) groups=1003(research_team),1004(sales_team)
analyst@3b53409cfca1:~$
```

Important: The lowercase -a option appends the supplementary group instead of replacing the user's existing supplementary group memberships. The uppercase -G specifies supplementary groups.

7. Task 4 — Delete the User

The employee leaves the organisation. Remove the researcher9 account, then remove the automatically created researcher9 group if it is no longer required.

Step 4.1 — Delete the user account

The lab may report that the researcher9 group was not removed because it is not the user's primary group. This is expected in this exercise.

Screenshot 5 — researcher9 user account deleted

```
userdel: group researcher9 not removed because it is not the primary group of user researcher9.
analyst@3b53409cfca1:~$ sudo userdel researcher9
userdel: user 'researcher9' does not exist
analyst@3b53409cfca1:~$ sudo groupdel researcher9
analyst@3b53409cfca1:~$
```

Step 4.2 — Delete the associated group

Screenshot 6 — researcher9 group removed

```
userdel: group researcher9 not removed because it is not the primary group of user researcher9.
analyst@3b53409cfca1:~$ sudo userdel researcher9
userdel: user 'researcher9' does not exist
analyst@3b53409cfca1:~$ sudo groupdel researcher9
analyst@3b53409cfca1:~$
```

8. Key Takeaways

- useradd creates a Linux user account.
- usermod changes account properties and group membership.
- The -g option assigns a primary group.
- The -a -G options append a supplementary group.
- chown changes file ownership.
- userdel removes a user account, while groupdel removes a group.